Manual

Extended Inspection Planning for Initial Sample Inspection

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1 Extended Inspection Planning for Initial Sample Inspection

Purpose

This component is intended to extend the functions of inspection planning by

- the assignment of gages,
- the entry of calculation formulas to calculate measured values from the results of other characteristics
- the definition of events to initiate inspections
- the print of inspection plans.

Implementation considerations

To the extent the initial sample inspection is to be performed on basis of VDA volume 2, 4th edition, and the related Initial sample inspection report is to be printed, use of this component is recommendable.

The same applies for the automatic measured value recording by means of an interface device since this requires the assignment of gages.

Integration

This component primarily serves the component "Inspection planning for initial sample inspection".

Features

The following functions are available:

- Indication of formula for the automatic calculation of the measurement result with reference to already collected measured values of previous inspection characteristics.
- Assignment of gages / gage groups with the possibility of using them for automated measured value recording by means of the HYDRA measurement data interface.
- Print of VDA initial sample inspection reports pursuant to VDA volume 2, 4th edition by using Word with the possibility of assignment to the document list of the related inspection request.
- Preparation of own initial sample inspection report forms if a license to create and manage Word forms is available.

2 Inspection Planning

Summary

Menu	Quality management → In-production inspection → Inspection planning Quality management → Goods receipt → Inspection planning Quality management → Goods issue → Inspection planning Quality management → Initial sample → Inspection planning Quality management → Gage management → Inspection planning
Transaction code	iplp
Function authorization	iplp

Utilization

Subject to the field of application, inspection planning is the most important function, as inspection plans are the basis for the generation of inspection orders and the resulting inspections. This means inspections in the area of goods receipt, production and goods issue as well as initial sample inspections and the calibration of test equipment and measurement equipment. As the requirements as regards inspection planning are nearly identical in all areas, the HYDRA inspection planning functions are also nearly identical for these areas.

In creating inspection plans, the user is supported in most cases by the master data catalogs. For this reason it is very important to conscientiously maintain this master data.

Integration

Inspection planning is a prerequisite to generating inspection requirements, the corresponding inspection steps and, as a result, the relevant inspections/calibrations.

The inspection plan in use is referenced in the generated inspection requirements. Moreover, some evaluations/reports also refer to inspection planning. Consequently, data of the same inspection plan number can be added to the basis of a control chart for inspection order characteristics. The same applies to the global control chart analysis, where special inspection plan filter fields may be used. The global control chart analysis does not take into account calibration data and, as a result, no filtered calibration inspection plans.

Another essential relation has been established to the production control plan. As no production control plan can be created without having generated inspection plans beforehand.

Prerequisite

Relevant master data needs to be edited/maintained to be able to create inspection plans. Which master data has to be maintained depends on the respective field of application. However, it is a fundamental prerequisite that articles/items and characteristics are maintained beforehand. Article groups also need to be created, provided that inspection plans based on article groups are created.

Selection criteria

The paragraph that follows shows some of the available selection criteria. Self-explanatory filter options are not listed.

Area

Selection list of the configured areas of in-production inspection, goods receipt and goods issue. By default, the following areas are available.

In-production inspection: Production

Goods receipt: Goods receipt

Goods issue: Goods issue

Initial sample: Initial sample inspection

Gage management: Gage management

Active

By checking the corresponding field, the inspection plan list can be restricted to active inspection plans. If this checkbox is not checked the list only shows inspection plans that are in the "in process" and "released" status. The third state of this checkbox (grayed out) shows all inspection plans. This is set by default.

Article number

Filters the article number with respect to inspection plans based on articles.

Article designation

Filters the article name with respect to inspection plans based on articles.

Article group

The article group tree can be opened by clicking the icon, in case inspection plans for article groups are to be filtered. A function to cancel and accept the entries made is provided.

Operation

Operation number

Customer number

Direct input or opening of the customer catalog and taking over of the number from the entry selected there.

Supplier number

Direct input or opening of the supplier catalog and taking over of the number from the entry selected there.

Manufacturer number

Direct input or opening of the manufacturer catalog and taking over of the number from the entry selected there.

Field descriptions

Inspection plan header

Area, inspection plan number, inspection plan index

Uniqueness of all existing inspection plans is secured by the "area", "inspection plan number" and "inspection plan index". The area may be selected. The inspection plan number and inspection plan index may be entered using alphanumeric characters. All three fields are mandatory fields.

The input of these three pieces of information must be unique, i.e. no other inspection plan may exist that already includes this information. By assigning a structured inspection plan number, it is possible to provide specific information. This information might be useful later during sorting. As an alternative, the inspection plan number may also be used synonymously with the article number or gage group number. The inspection plan index corresponds to the inspection plan version. If an existing inspection plan version is to be modified, yet it cannot simply be edited because it has already been used to generate inspection orders, it is recommended to copy the original inspection plan and then modify the inspection plan index (e.g. incrementing it by 1). The inspection plan number should be maintained, if possible, to facilitate later evaluations/reports (e.g. for displaying control charts applying to several orders, in which case the inspection plan versions on which the evaluation is based differ while the inspection plan number is identical).

Inspection plan type

The inspection plan types "article inspection plan" and "group inspection plan" are distinguished. The "article inspection plan" type allows for an article to be assigned from the article master data for which this inspection plan is to be created. The "group inspection plan" option has to be selected to create an inspection plan for article groups. This allows for an article group to be selected. The selected article group is displayed in a separate field within the tree structure.

Article

Input of the article number. Provided that it is known, it may be entered directly. Otherwise, the article catalog can be opened and the requested article can be identified and taken over using the filter and sort criteria provided there. By selecting an article, the drawing issue number, article designation, customer article number and the drawing number are taken over from the master data record and displayed in the corresponding fields.

The customer article number is not displayed within the goods receipt inspection planning.

Nothing may be entered here if the field “article number” or “article group” is shown in gage management.

Drawing issue number

The drawing issue number can be entered directly, just as it is the case for the article number.

In case the article number and the drawing issue number are entered directly, the master data record is determined on the basis of this information while saving and the article designation, customer article number as well as the drawing number are displayed.

The article is uniquely identified by the combination of article and drawing issue number. It is not obligatory to define and use a drawing issue number.

Nothing may be entered here if the “drawing issue number” field is shown in gage management.

Operation assignment

One of the two assignment types "an inspection plan for each OP" and "an inspection plan for all OPs" may be selected.

By assigning operations, it is defined whether the inspection plan to be created should include the characteristics of different operations ("one inspection plan for all OPs") or whether the inspection plan should only include the characteristics of one operation. The fields "operation" and "operation designation" are shown if the "an inspection plan for each OP" option is selected. In any other case, operation details are assigned in the area of inspection plan characteristics. The option "one inspection plan for all OPs" should be selected as far as possible. This allows, for example, printing out of all inspection characteristics of an article, although it has several operations.

The option “one inspection plan for all OP“ should be used in areas that are not assigned to a production order. This provides the advantage that the “fictious” operation only has to be entered once in the inspection plan header, instead for every inspection plan characteristic.

However, the system has to be configured accordingly by MPDV consultants to be able to use this inspection plan variant.

Operation, operation designation

If operations are used, only one of the two fields "operation" or "operation designation" needs to be filled out. Only the operation number has to be entered in the "operation" field if an inspection requirement/inspection order is to be generated automatically by logging an operation of a production order on.

It has to be taken into consideration that these fields are used as search criteria when inspection orders are generated at a later point in time. Provided that only the designation is entered for the operation within the inspection plan, exclusively this information needs to be provided when inspection orders are generated.

A “fictious“ operation has to be assigned (e.g. 9999) even in areas that are not assigned to a production order (e.g. in goods receipt or in gage management). This is necessary to generate inspection steps at a later point in time.

Released/active

Shows whether the inspection plan is "released" and/or "active". If the inspection plan is released or active the corresponding checkboxes are checked. An inspection plan is released and activated, i.e. its status is changed, only by using the corresponding toolbar functions. An inspection plan has to be released before it can be activated.

Only active inspection plans are taken into consideration, when inspection orders are generated at a later point in time.

Released by / on

Shows the HYDRA user who has released it. The release date is displayed additionally.

Valid from / until

Instead of an "unrestricted" activation (using the toolbar), a validity period may be entered here. This period is then taken into account when the inspection order is generated. Yet activation for a certain period means that the user has no clear overview of currently valid inspection plans, and it is therefore recommended to use the "global/unrestricted" activation option using the toolbar if possible. If activated via the toolbar functions, the system carefully monitors whether an active inspection plan exists for the specified article and checks whether this plan includes the same drawing issue number, customers and suppliers. If this is indeed the case, the previously active inspection plan is automatically deactivated.

Type of inspection

Inspection based on characteristics or pieces

When it comes to an inspection based on characteristics and, say a sample size of 5, the characteristic is checked completely first. Whereas with the inspection based on pieces, the characteristic is changed every time a measured value is collected, as items are checked entirely.

Action

Create inspection step or create and immediately release inspection step

Only inspection steps that have been released can be checked.

Customer / supplier / manufacturer

If a customer / manufacturer or supplier is selected this inspection plan only applies for the selected company. Consequently, the customer / manufacturer / supplier becomes the key field of an active inspection plan. In this case, the inspection requirement must also include a company entry in order for this inspection plan to be used for the generation of the inspection requirement.

Dynamic modification type

Characteristic-related, batch-related or none

The authorization “iriscp.dynamic“ is required to show these fields.

Batch-related: In this case, it is defined on the level of inspection plan headers which transitional definition is to be used for the dynamic modification. In addition, the initial inspection severity is also defined on the level of inspection plan headers. Now only the reference to the dynamic modification norm is required on the level of inspection plan characteristics. In this context, however, the selection is restricted to those dynamic modification norms referring to the same inspection severity definition as the assigned transitional definition.

Characteristic-related: In this case, the transitional definition, initial inspection severity as well as the dynamic modification norm are referenced on the level of inspection plan characteristics. In this context, however, the inspection severity definition of the assigned transitional definition has to match the assigned dynamic modification norm.

None: The "dynamic modification" function has not been enabled for this inspection plan.

Transitional definition

The authorization “iriscp.dynamic“ is required to show these fields.

This field is only available if a dynamic modification relating to batches or characteristics is selected. It provides the possible inspection severities and controls switching between the inspection severities.

Initial inspection severity

The authorization “iriscp.dynamic“ is required to show these fields.
This field is only available if a dynamic modification relating to batches or characteristics is selected. It defines which inspection severity is used for checking the first goods received according to the basics for the dynamic modification history.

Initial sample form

The different form types available for initial sampling can be chosen here. The form type “VDA volume 2, 4th edition“ is only supported at the moment. Subject to the selected form type, a corresponding list of categories is shown in the inspection plan characteristics.

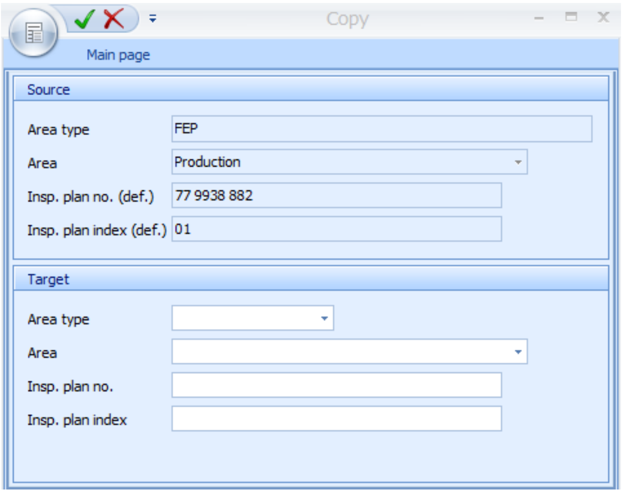
Editing functions

The key fields "area", "inspection plan" and "inspection plan index" cannot be changed in the editing mode.

Toolbar

Copy

The below dialog opens to copy an inspection plan:



The target area type and target area may be entered here. Normally, the user should select an area that is identical to that of the source inspection plan. The new inspection plan number and inspection plan index are to be entered afterwards. In case a new version is generated from an existing inspection plan, normally the same inspection plan number is used and only the inspection plan index is changed.

Activate

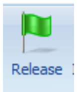
Function authorization: iplp.activate

Makes the inspection plan status "active“.

Deactivate

Function authorization: iplp.deactive

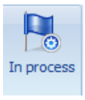
Puts an inspection plan that is in the "active" status back to the "released" status.



Release

Function authorization: iplp.release

Puts an inspection plan that is in the "in process" status to the "released" status.



In process

Function authorization: iplp.inprocess

Puts an inspection plan that is in the "released" status in the "in process" status.

"Print" detail application

Function authorization	iplp.print

The print dialog of the inspection plan header opens a list of available reports. These are Word forms. The potential content of these forms is determined by the Web services that are available in the respective context. The form entries, i.e. the content of the form list of the corresponding print dialog, are defined within the master data of quality management. This is where the basis for new forms is established and the corresponding form properties are defined. A corresponding license is required to be able to change the forms with respect to content and design.

Toolbar - print

There are no other special function buttons in addition to the standard functions.

"Inspection plan characteristics" detail application

The detail application of inspection plan characteristics is nearly identical to the master data of characteristics application. For this reason, reference is made only to additional features in the following.

Go to

For further information on the definition of characteristics, please refer to the functional description of the document entitled MOC_CharacteristicsQM.

On the level of inspection plan characteristics the corresponding inspection plan characteristics are assigned to the previously defined inspection plan header. Assignments are performed by way of initial data creation using the characteristics catalog and taking over the characteristic selected there. By accepting the characteristic, all master data entries are copied to the inspection plan characteristic. Every (copied) detail may be changed and / or complemented afterwards. The characteristic designation is often complemented to define it in more detail.

It is also possible to create a characteristic not included in the characteristics catalog. However, this is recommended only in exceptional cases, since all analyses (e.g. failure mode analysis) are based on characteristics found in the catalog. It is therefore preferable to properly maintain the characteristics catalog.

Different properties and settings can be defined, before specific characteristic data is complemented.

Field descriptions

Inspection plan characteristics

OP sequence

The operation sequence number (OP sequence/AFO) determines the order in which the inspection is later performed. Entries must be unique. In the ideal case, operation sequence numbering should be assigned in steps of 10. This provides the option of later inserting a new characteristic between two existing ones.

Characteristics number

Number of the characteristic selected from master data

Characteristic type

Decides whether the inspection is based on the collection of measured values (variable) or on the specification of the number of detected failures (attributive). When it comes to the attributive inspection, the decision is often only based on the "pass" or "fail" results. Further characteristic types are the inspection chart and the information characteristic. The information characteristic has only been designed to display a document while the inspection is running. Subject to the input type, the lower area of the dialog provides corresponding sampling schemes.

Inspection result base

Decides whether all samples or only the last sample is used to determine the inspection result (pass/fail).

Mandatory inspection

If this checkbox is activated, at least one measured value has to be entered for this characteristic, before an inspection order can be completed with this characteristic.

Calculate characteristic

Identifies a characteristic to be calculated

Formula

Further details can be taken from the manual dealing with the CAQ master data characteristics.

Initial sample creation

Shortlist of categories for initial sampling. The contents depend on the form type selected in the inspection plan header. Within the resulting initial sample report, the inspection plan characteristics are summarized by category and a separate page is printed for each group.

Copy characteristic

It may be defined here which characteristic is to be copied from an initial sample inspection plan to a production plan, for example. Only those characteristics are copied that have been marked relevant in this field.

Details

Defines whether - with respect to this characteristic - the contents of the "details" tab are defined within the inspection plan or whether they have to be determined from the available master data characteristic or an entry from the specification list when inspection requirements are later generated on the basis of this inspection plan. If the "from characteristic catalog" option is selected the "details" tab is hidden. The "from characteristics catalog" option is reasonable if it is a characteristic the specifications of which are to be defined identically for several articles (even applicable for several article groups). In this case, specifications can be defined centrally within the master data. This reduces the input and update efforts considerably.

Specifications

Defines whether - with respect to this characteristic - the contents of the "specification" tab are defined within the inspection plan or whether they have to be determined from the available master data characteristic or from a specification list entry, when inspection requirements are later generated on the basis of this inspection plan. The tabs "specifications", "chart 1", "default values chart 1", "chart 2" and "default values chart 2" are hidden if the option "from list" or "from characteristic catalog" is checked. The "from list" option is only reasonable if inspection plans for article groups are in use. The specification list is used for an inspection plan for article groups if the specifications vary from item to item for certain characteristics.

Dynamically modified

The authorization “iriscp.dynamic“ is required to show these fields.
This field is only available if a dynamic modification relating to batches or characteristics has been activated in the inspection plan header. This option allows for definitions to be made that a characteristic is not to be modified dynamically, although the dynamic modification option is selected in the inspection plan header.

Transitional definition

The authorization “iriscp.dynamic“ is required to show these fields.
This field is only available if a dynamic modification relating to batches or characteristics has been activated in the inspection plan header. The transitional definition that can be selected provides the possible inspection severities and controls switching between these different inspection severities. For the dynamic modification based on batches, the transitional definition is assigned in the inspection plan header and is not available on the level of inspection plan characteristics in this case.

Initial inspection severity

The authorization “iriscp.dynamic“ is required to show these fields.
It defines which inspection severity is used for checking the first goods received according to the basics for the dynamic modification history. Only the inspection severities of the previously selected transitional definition are available. For the dynamic modification based on batches, the transitional definition is assigned in the inspection plan header and is not available on the level of inspection plan characteristics in this case.

Norm

The authorization “iriscp.dynamic“ is required to show these fields.
This field is only available if a dynamic modification relating to batches or characteristics is activated in the inspection plan header and if this characteristic is actually characterized by being relevant to dynamic modification. The dynamic modification norm to be used is assigned from master data.

Inspection level

The authorization "iriscp.dynamic" is required to show these fields.

This field is only available if a dynamic modification relating to batches or characteristics is activated in the inspection plan header and if this characteristic is actually characterized by being relevant to dynamic modification. An inspection level can only be selected if the assigned norm corresponds to DIN ISO 3951 or DIN ISO 2859.

AQL

The authorization "iriscp.dynamic" is required to show these fields.

This field is only available if a dynamic modification relating to batches or characteristics is activated in the inspection plan header and if this characteristic is actually characterized by being relevant to dynamic modification. An AQL value can only be selected if the assigned norm corresponds to DIN ISO 3951 or DIN ISO 2859.

Method

The authorization "iriscp.dynamic" is required to show these fields.

This field is only available if a dynamic modification relating to batches or characteristics is activated in the inspection plan header and if this characteristic is actually characterized by being relevant to dynamic modification. A method can only be selected if the assigned norm corresponds to DIN ISO 3951.

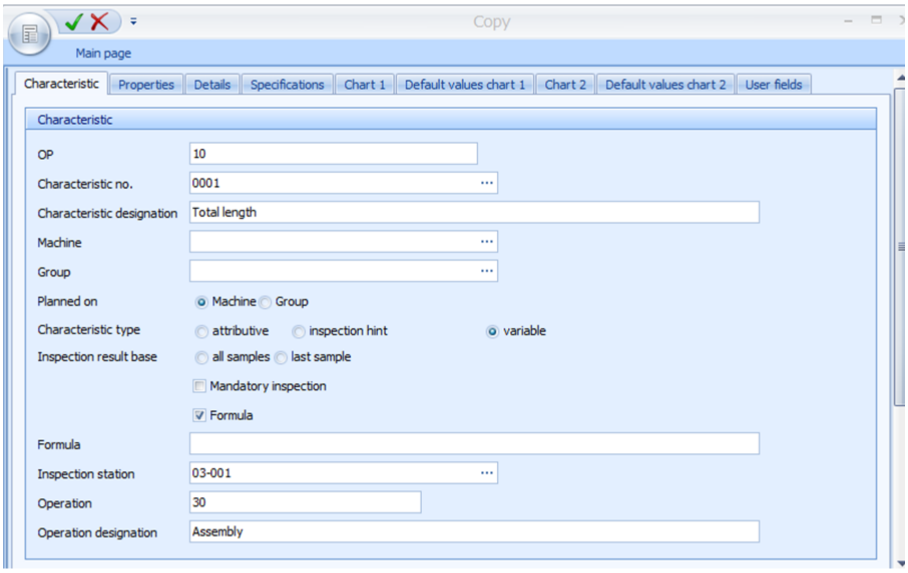
Go to

The fields of the respective tabs correspond to those of the characteristics master data. Further information on these fields can be gathered from the functional description of the document entitled MOC_CharacteristicsQM.

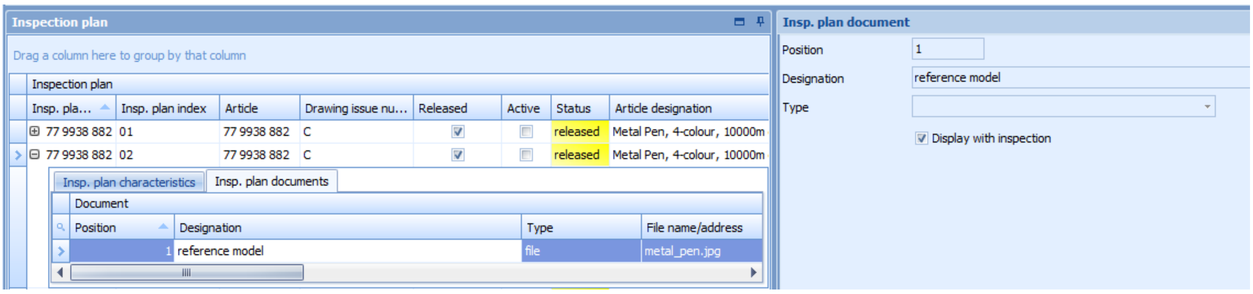
Inspection plan characteristics - toolbar

Copy

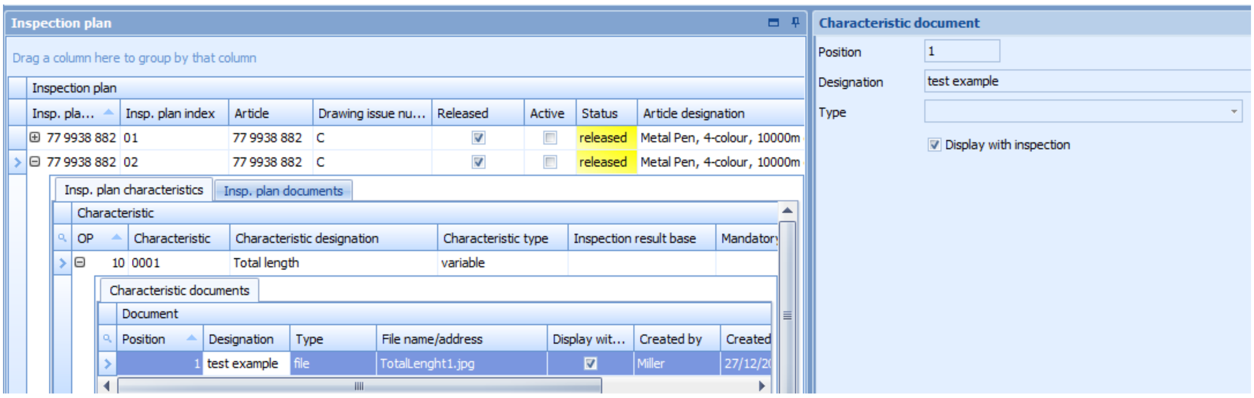
Data of the selected characteristic is opened in insert mode to be able to copy inspection plan characteristics. All fields can be changed. An operation sequence number that does not yet exist in this inspection plan has to be assigned to be able to save it.



"Inspection plan documents“ and "documents of inspection plan characteristics“ detail applications

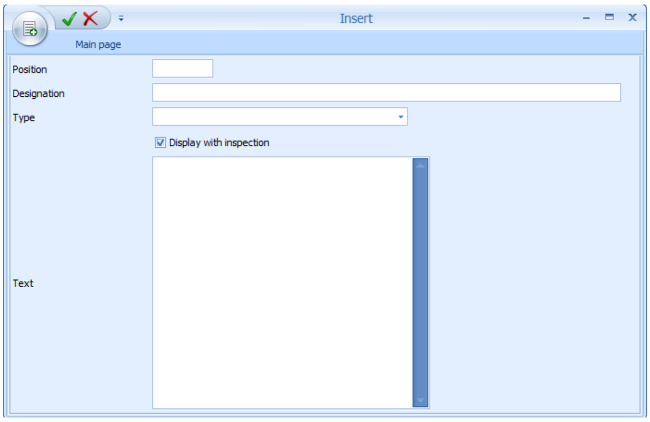


The above screenshot shows how an inspection plan document is assigned



The above screenshot shows how a document of an inspection plan characteristic is assigned

Provided that the "inspection plan documents" tab or the "characteristic documents" tab has been activated in the master detail grid, as many documents as required may be assigned to each inspection plan and each inspection plan characteristic. By enabling these tabs, the toolbar offers corresponding buttons to edit the documents.



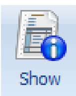
When documents are assigned, all formats registered by Windows are provided. Consequently, it is possible to assign simple documents (e.g. written in Word), drawings of any format and videos. However, the corresponding programs that are able to display the required formats have to be installed. In this context, the documents are opened by the program that has been linked in Windows.

The file types are: "FILE", "URL" and "Text". The file name including path may be entered manually with the "file" type. The "URL" file type enables access to the Internet or Intranet. The third file type "Text" allows for text to be entered directly.

A designation may be assigned to each defined document. Moreover, it may be determined in which order the documents are to be listed. The "position" field is used for this purpose (numeric input). The specifications made within this list must be unique. In addition, the checkbox "display with inspection" determines whether or not the document may be shown during the inspection process.

"Inspection plan documents“ and "documents of inspection plan characteristics" - toolbar

In addition to the standard features, there is also a button to display the documents.



Show documents

If a document link is defined this button opens and shows this document. However, a program, which is able to visualize the linked file type, has to be installed on the PC.